TRACE LAB
Trustworthy AI, Reasoning Algorithms, and Causal lEarning

Causality and Large Language Models: A Survey of System Design, Capabilities, Trustworthiness, and Applications

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Abstract

Large language models (LLMs) have advanced rapidly in recent years, demonstrating capabilities in reasoning, generation, and planning across diverse applications. Causality, a fundamental aspect of human intelligence, enables reasoning about cause-and-effect relationships and has contributed to advances in machine learning. Recent research has increasingly explored the use of causal methods to improve LLM capabilities, system design, and trustworthiness. However, existing studies span diverse methods and tasks, highlighting the need for a systematic understanding of how causality contributes to different aspects of LLMs. In this survey, we examine the role of causality in four main aspects of LLMs: system design, capabilities, trustworthiness, and applications. We introduce a taxonomy to organize existing approaches and examine how causal principles are used across these directions. We further review evaluation methods for all four aspects from a causal perspective. Finally, we identify key challenges and open research directions to guide future research on causality for LLMs.

1 Introduction

Large language models (LLMs) have advanced rapidly in recent years, demonstrating strong performance across a wide range of applications (Achiam et al., 2023; Grattafiori et al., 2024). Despite these advances, LLMs continue to face challenges in generation, reasoning, planning, and reliable deployment across real-world applications and downstream tasks due to their reliance on spurious correlations (Bynum and Cho, 2025; Han et al., 2025; Zan et al., 2026; Chai et al., 2025; Lyu et al., 2024).

Causality is a fundamental aspect of human intelligence that enables us to understand cause-and-effect relationships and reason about hypothetical interventions (Lake et al., 2017). Causality has long contributed to advances in machine learning Kaddour et al. (2025). Motivated by these advances, researchers have increasingly explored the integration of causal principles into LLMs (e.g., Chernozhukov et al. (2018); Guo et al. (2023); Nguyen et al. (2025)). LLM research spans four main aspects: system design, capabilities, trustworthiness, and applications. System design focuses on model development and LLM-based architectures; capabilities include generation, reasoning, planning, and decision-making; trustworthiness concerns fairness, faithfulness, interpretability, security, and alignment to ensure reliable deployment in real-world, high-stakes applications; and applications focus on the use of LLMs in downstream tasks. These aspects motivate a systematic examination of how causality can contribute to different areas of LLM research.

In this survey, we present a systematic overview of research at the intersection of causality and LLMs. We examine the use of causality across four main aspects of LLMs: system design (Section 2), capabilities (Section 3), trustworthiness (Section 4), and applications (Section 5). We introduce a taxonomy that organizes existing approaches according to their tasks, subtasks, and methods. Figure 1 presents an overview of the proposed taxonomy, and a more detailed version is presented in Appendix B. We further review evaluation

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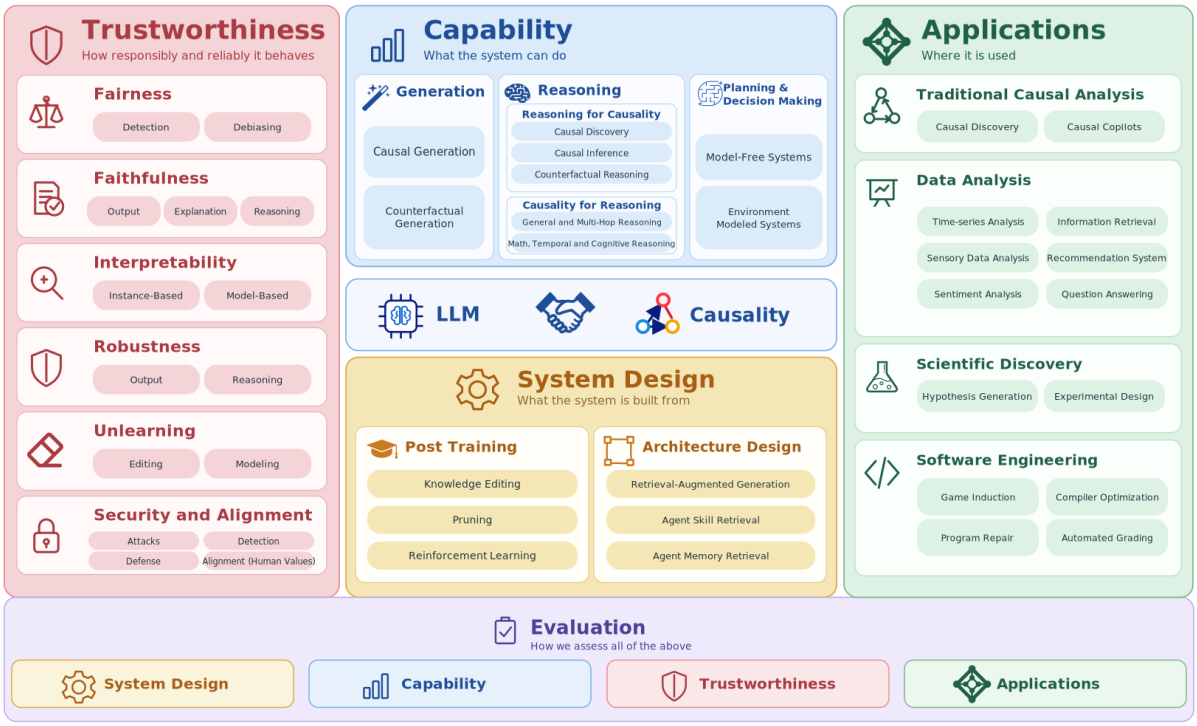


Figure 1: Overview of the proposed taxonomy, examining the role of causality across four main aspects of LLMs: system design, capabilities, trustworthiness, and applications, along with causal evaluation methods across these aspects.

methods for all four aspects from a causal perspective (Section 6). Finally, we discuss key challenges and open research directions at the intersection of causality and LLMs (Section 7).

Differences from Existing Surveys. The survey most closely related to ours is (Liu et al., 2025c), which examines both the use of LLMs for causal tasks and the use of causality to improve LLM capabilities and trustworthiness. However, its coverage of causal methods for LLM system design, downstream applications, and evaluation is limited. Other surveys focus on specific directions. (Zhang et al., 2023) discuss the feasibility and opportunities of integrating causality and LLMs, whereas (Wu et al., 2024a; Frauen et al., 2026) examine causal methods across LLM development and evaluation. Several surveys focus on specific causal capabilities, including commonsense causal reasoning (Cui et al., 2024), causal inference (Ma, 2025), causal reasoning and evaluation (Yu et al., 2025a), causal uncertainty (Cui et al., 2025), and causal reasoning enhancement (Li et al., 2025c). Other studies examine LLMs for causal discovery (Wan et al., 2025) and causal multi-agent systems (Bazgir et al., 2025). In contrast, our survey systematically examines how causality contributes to four main aspects of LLM research: system design, capabilities, trustworthiness, and applications. We introduce a taxonomy that organizes existing studies by tasks, subtasks, and methods, and examine causal evaluation strategies across these aspects. A detailed comparison with existing surveys is presented in Table 2, Appendix C.

2 Causality and System Design

In this section, we discuss causal post-training methods for improving LLM performance, as well as the use of causality in designing more complex LLM systems.

2.1 Post Training

LLM architectures can be enhanced through post-training methods to refine model's knowledge, behavior, and efficiency (Ouyang et al., 2022; Meng et al., 2022; Frantar and Alistarh, 2023; Wang et al., 2020). We

discuss how causal approaches improve such training paradigms.

Reinforcement learning (RL) traditional RL methods such as GRPO (Shao et al., 2024) treat each candidate response as conditionally independent, overlooking semantic dependencies or interactions among the candidates. Modeling the relations between the input query, candidate responses, and final outcome with a Structural Causal Model (SCM), GCPO guides the policy updates towards responses that are individually useful and structurally coherent with the group (Gu et al., 2026).

Knowledge editing selectively updates model components without full-retraining (Meng et al., 2022). To overcome the epistemic dissonance, which occurs due to isolated fact injection and recursive error accumulation, (Li et al., 2026f) propose to distill the causal reasoning trajectories with knowledge transition with asymmetric on-policy distillation. (Meng et al., 2022) identify critical layers with causal tracing and adaptively update them based on their causal efficacy, reducing the recursive error accumulation.

Pruning removes redundant or unnecessary model parameters (Frantar and Alistarh, 2023) to improve model efficiency while preserving performance. However, correlations among parameters may cause conventional pruning methods to remove causally important parameters. To address this, (Sheth et al., 2026a) use causal analysis to identify and preserve parameters that have a causal effect on model predictions, even when they are correlated with less relevant parameters. In another line of research, (Guo et al., 2025c) propose identifying tokens that confound the causal relationship between instructions and responses by comparing their gradients with those from a stronger teacher model. These tokens are then pruned to reduce spurious correlations in long-context reasoning.

2.2 Architecture Design

To extend the capabilities of LLMs and address their limitations in complex tasks, LLM-based architectures like retrieval-augmented generation and LLM agents have been developed. We discuss the role of causality in improving these architectures.

Retrieval-Augmented Generation (RAG) enhance LLMs performance by providing relevant external information for informed generation (Lewis et al., 2020), although similarity-based retrieval may retrieve causally irrelevant information. CAUSALRAG uses document-level causal graphs and discovers causal paths among the nodes to guide the retrieval process (Wang et al., 2025a), while CAUSALRAG2 (Wang et al., 2026b) extends this to hierarchical knowledge graphs with causal gates to filter for required documents causally. To reduce graph construction costs HCG-RAG (Saouda et al., 2026) constructs an outcome-oriented causal graph that decreases the required number of nodes which improves the graph construction efficiency while preserving causal relationships relevant to the given query.

LLM Agents use memory and tools for complex tasks (Park et al., 2023; Wang et al., 2024). Similarity-based memory retrieval methods can overlook contextual relevance, leading to irrelevant or misleading information. CAUSAL MEMORY INTERVENTION (Srivastava, 2026) estimates the causal effect of retrieved memories on task performance by comparing no-memory, with-memory, and perturbed-memory conditions, and filters out irrelevant or harmful memories. ACTMEM (Zhang et al., 2026b) represents conversation history as a causal graph and uses counterfactual reasoning to predict the consequences of retrieved facts, enabling the retrieval of causally relevant information. Traditional skill-retrieval methods often require long contexts, treat skills independently, or rely on accurate relevance estimates. To address these limitations, CASKG (Li et al., 2026h) constructs a directed skill graph and refines its edges through task-specific counterfactual perturbations. This enables agents to retrieve skills while preserving their procedural dependencies and execution order.

3 Causality and Capability

LLM capabilities refer to the ability of the models to perform tasks across different settings using reasoning, generation, planning, and decision-making. We discuss how causality is utilized to improve three major LLM capabilities: generation, reasoning, and planning & decision-making.

3.1 Generation

LLM generation refers to the ability to produce coherent outputs based on a given input (Brown et al., 2020). We categorize the synergy of LLM generation and causality as Causal Generation and Counterfactual Generation.

3.1.1 Counterfactual Generation

Counterfactual generation produces alternative versions of observed scenarios by intervening in specific conditions while preserving the relevant context and altering outcomes (Pearl, 2009).
Prompting: Prompt-based methods generate counterfactuals through targeted textual span extraction (Chen et al., 2023) or neuromorphic-inspired pattern separation and completion to improve commonsense consistency (Miao et al., 2024).
Training: Training-based approaches improve generation quality through counterfactual data augmentation and preference optimization (Zhang et al., 2025b) or classifier-guided selection of effective label-flipping counterfactuals (Nguyen et al., 2026a).
Architecture: At the architecture level, SCMs with Gumbel-Max distributions (Gumbel, 1954) enable token-level counterfactual generation conditioned on factual LLM outputs (Chatzi et al., 2024; Ravfogel et al., 2025), although robustness and faithfulness remain challenging.

3.1.2 Causal Generation

Causal generation refers to the ability of LLMs to generate outputs under a causal model and specific interventions that propagate the causal effects to downstream generation (Bynum and Cho, 2025). For example, SD-SCM (Bynum and Cho, 2025) combines user-provided DAG with LLM to generate data consistent with user-specified DAG.

3.2 Reasoning

LLM reasoning refers to the ability to process information and draw conclusions through logical steps. We categorize its synergy with causality into Reasoning for Causality (Section 3.2.1) and Causality for Reasoning (Section 3.2.2).

3.2.1 Reasoning for Causality

Based on the ability of LLM to perform reasoning, studies have also investigated the ability of LLM to perform causal reasoning (Kiciman et al., 2024; Chen et al., 2024a; Gao et al., 2023).
Causal Discovery identifies causal relationships among variables from observational data, to recover the underlying causal structure as a DAG or an SCM (Kiciman et al., 2024).
Prompting methods infer pairwise causal relationships (Kiciman et al., 2024), with extensions improving consistency through triplet-based causal ordering (Vashishtha et al., 2025), reducing query complexity from quadratic to linear through breadth-first search (Jiralerspong et al., 2024), and incorporating causal adjacency matrices with collider detection (Bagheri et al., 2025) or structured knowledge graphs (Sun et al., 2025a) to guide causal discovery.
Training approaches improve out-of-distribution (OOD) generalization by incorporating human-derived causal priors into attention mechanisms through Re-Attention (Han et al., 2025).

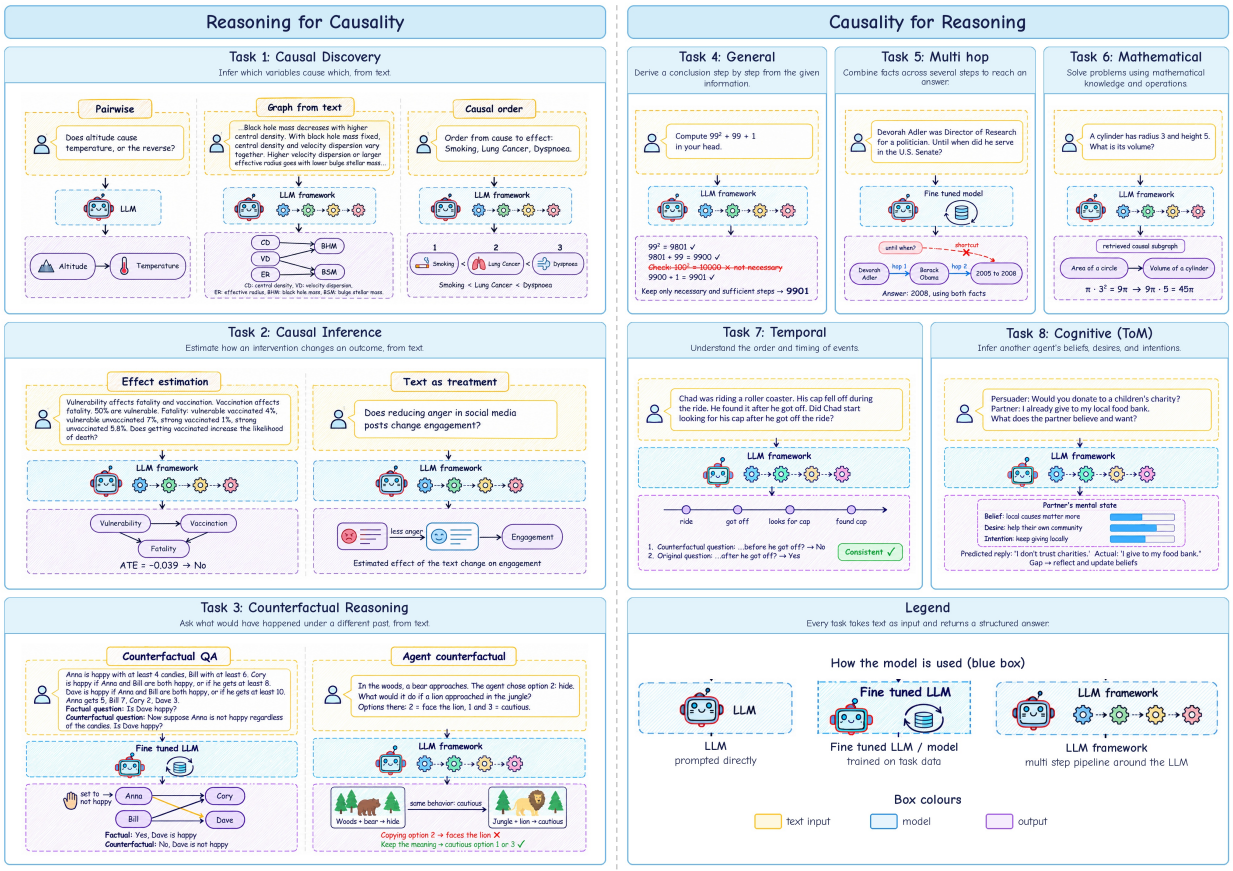


Figure 2: Illustration of tasks in reasoning for causality using LLMs.

Causal Inference estimates the effect of intervening on a variable X (cause or treatment) on an outcome Y (effect). A fundamental task in causal inference is causal effect estimation, which quantifies the effect of interventions on outcomes.

Prompting approaches guide causal effect reasoning through structured chain-of-thought (CoT) (Jin et al., 2023) or improve auditability using target-aware causal graphs, causal chain verification, and evidence aggregation (Lan et al., 2026)

Training methods address high-dimensional textual data through LLM-based nuisance estimation within the Double Machine Learning framework (Chernozhukov et al., 2018) for confounding adjustment (Veljanovski and Wood-Doughty, 2024) and LLM-generated textual interventions with domain-adversarial learning to estimate average and conditional treatment effects (Guo et al., 2025b). Alternatively, axiomatic training uses diverse synthetic causal examples to improve reasoning generalization (Vashishtha et al., 2024).
Counterfactual Reasoning reasons about the effects of hypothetical interventions against factual observations Pearl (2009).

Prompting Prompt-based approaches follow the abduction-action-prediction framework (Pearl, 2002), to infer latent factors, apply hypothetical interventions, and predict alternative outcomes (Pona et al., 2025).

Training Training-based methods improve counterfactual question answering through factual and counterfactual feedback combined with supervised fine-tuning and preference optimization to enhance generalization (Hüyük et al., 2025).

LLM Aspect	Benchmarks	Datasets
Causality and System Design	(Chen et al., 2026a)	(Luan et al., 2026; Shah, 2026)
Causality and Capability	(Jin et al., 2023; Chen et al., 2024b; Sheth et al., 2025; Yu et al., 2025a; Li et al., 2026e; Du et al., 2026; Zečević et al., 2023; Leban and Sun, 2026; Acharya et al., 2025; Chen et al., 2026a, 2024a; Xu and Fu, 2026; Chi et al., 2024; Zhang et al., 2025d; Romanou et al., 2023; Wang, 2024a; Wang et al., 2026e; Liu et al., 2024b; Sawarni et al., 2026; Sheth et al., 2024; Babakov et al., 2025; Ding et al., 2026; Chen et al., 2026c; Wu et al., 2025; Joishy et al., 2026; Sonkar et al., 2024; Nguyen et al., 2024; Ong et al., 2025; Bynum and Cho, 2025; Chen et al., 2026d)	(Miliani et al., 2025; Jin et al., 2024; Astorga et al., 2026; Liu et al., 2024b; Li et al., 2026b; Roemmele et al., 2011; Du et al., 2022; Tandon et al., 2019; Toroghi et al., 2026; González and Nori, 2024; Carro et al., 2025; Scutari, 2010a; Sachs et al., 2005; Mooij et al., 2016; Erdős and Rényi, 1959; Barabási and Albert, 1999; Frohberg and Binder, 2022; Vashishtha et al., 2026; Huang et al., 2024c)
Causality and Trustworthiness	(Xu and Fu, 2026; Liu et al., 2023c; Wang and Liu, 2026; Han et al., 2026; Adewuyi et al., 2026; Chen et al., 2026b; Mo et al., 2026; Karvonen et al., 2026; Chen et al., 2025c; Toker et al., 2026; Zaman and Srivastava, 2025; Mittal and Arike, 2026; Arora et al., 2024; Huang et al., 2024a; Mueller et al., 2025; Ye et al., 2026)	(Carro et al., 2025; Hong and Roth, 2026; Shah, 2026; Mayilvaghanan et al., 2026; Parziale et al., 2026a; Xie et al., 2026; Zhang et al., 2025e)
Causality and LLM Applications	(Leban and Sun, 2026; Acharya et al., 2025; Liu et al., 2024b; Sawarni et al., 2026; Sheth et al., 2024; Chen et al., 2026d)	(Liu et al., 2024b)

Table 1: Benchmarks and datasets for causal evaluation of large language models, grouped by the four LLM aspects in the taxonomy. Each cell cites the studies that introduced the resources.

3.2.2 Causality for Reasoning

LLMs have showcased capability of other reasoning tasks. In this section, we focus on those with the use of causality.

General and Multi-Hop Reasoning General reasoning is the ability of LLMs to analyze information and derive conclusions, while multi-hop reasoning integrates multiple pieces of evidence through intermediate reasoning steps (Jiang et al., 2022). To address irrelevant reasoning steps and shortcut dependencies in CoT reasoning (Wei et al., 2022), causal approaches estimate step importance through probabilities of necessity and sufficiency (Pearl, 2002) using counterfactual replacements, subsequently improving reasoning through fine-tuning or in-context learning (Yu et al., 2025b). Similarly, multi-hop reasoning can be modeled as a DAG, with counterfactual inference distinguishing shortcut paths from valid reasoning dependencies to guide training (Guo et al., 2023).

Mathematical, Temporal, and Cognitive Reasoning Specialized mathematical reasoning benefits from causal discovery over mathematical knowledge and question-solution pairs, with relevant causal subgraphs retrieved to guide problem-solving (Zan et al., 2026). Temporal reasoning uses counterfactual question variations and predictions aggregation to improve consistency under lexical perturbations (Kim and Hwang, 2025). For cognitive reasoning, particularly, Theory of Mind (ToM), counterfactual reflection supports reasoning about alternative mental states TOM-AGENT(Yang et al., 2025), while causal tracing and activation steering target ToM-related representations to improve social reasoning (Li et al., 2026d). Additionally, SCM-based opponent modeling captures dependencies between opponent's states and actions to improve behavioral prediction in dynamic interactions (Cao et al., 2026)

3.3 Planning & Decision Making

Planning organizes a sequences of actions to achieve a goal, while decision making selects a specific action for a goal (Yao et al., 2023b,a). Although planning & decision making are an application of reasoning capability, we discuss this as a separate section due to its importance to enable LLMs for goal-directed behavior and autonomous task execution. We divide the planning & decision-making tasks into two main

categories: Model-Free Systems and Environment Modeled Systems.

3.3.1 Model-Free Systems

Model-Free systems enable LLMs to plan and make decisions without explicitly learning or generating a predictive model of the environment dynamics, instead relying on observations and environmental feedback from the environment (Yao et al., 2023b; Shinn et al., 2023).
Decision Making: ARGORA (Jin et al., 2026) uses argumentation graphs and counterfactual inference to quantify influential factors in multi-agent decisions.
Planning: CAUSALMACE (Chai et al., 2025) uses causal inference to refine task dependencies under game and task constraints, guiding coordinated multi-agent planning.

3.3.2 Environment Modeled Systems

Environment modeled systems learn a predictive model of the environment capable of helping agents simulate the effects of actions on future states and support planning & decision making (Hao et al., 2023; Guan et al., 2023; Tang et al., 2024).
Decision Making: SCM is used to represent state-action dependencies, enabling adaptive decision making through environmental interventions and model updates (Chen et al., 2025b).
Planning: SCM-based methods support causal action dependency discovery (Nguyen et al., 2025), counterfactual refinement of actions in stochastic environments (Fu et al., 2026) are two lines of research. Additionally, causal representation learning (Schölkopf et al., 2021) enables the construction of causal world models for causally informed planning (Gkountouras et al., 2025).

4 Causality and Trustworthiness

LLM trustworthiness refers to the extent to which LLM behavior is fair, faithful, interpretable, robust and safe while aligning with human preferences and ethics (Liu et al., 2023b; Huang et al., 2024b). With the synergy of causality, the trustworthiness of LLMs can be enhanced by learning the underlying behavior of the model, mitigating spurious correlations, and enabling a causal analysis of the LLM's predictions and decisions. In this section, we discuss the role of causality for different aspect of trustworthiness.

4.1 Fairness

LLMs have demonstrated bias and reproduce stereotypes in their generated outputs (Parrish et al., 2022; Dhamala et al., 2021). For detection, counterfactual fairness testing evaluates sensitivity of the model to sensitive attributes through stereotype-aware prompt generation (Parziale et al., 2026b), while causal interventions on inputs of agentic evaluator identify sources of bias (Chen et al., 2025a). For debiasing a black-box model, a prompting framework uses *dual direction priors* along with *selective mechanisms* to mitigate biased generation (Li et al., 2025a). Training-based methods address spurious correlations and confounding through counterfactual data augmentation (Madhavan et al., 2023), interventional reinforcement learning (Xia et al., 2024), causal-contrastive learning with fairness-sensitive objectives (Feng et al., 2026), and adversarial learning of de-biased steering representations (Chu et al., 2024).

4.2 Faithfulness

Faithfulness assess whether LLM generated outputs are grounded to the source information and whether their reasoning and explanations accurately reflect model behavior (Turpin et al., 2023; Maynez et al., 2020). Recent studies focus on the faithfulness of LLMs explanations from 3 major perspectives; output, explanation, and reasoning faithfulness.
Output faithfulness For output faithfulness, counterfactual probing and iterative refinement assess sensitivity to factual perturbations (Feng, 2025; Narendra and Chatterjee, 2026), while causal graph-guided fine-tuning and causal-effect based abstention aim to mitigate hallucinations (Li et al., 2026g; Nguyen et al., 2026b).

Explanation faithfulness For explanation faithfulness, causal concept interventions quantify the influence of explanatory factors (Matton et al., 2025), while studies reveal the unreliability of self-generated counterfactual explanations (Madsen et al., 2024; Mayne et al., 2025). Alternatively, counterfactual simulatability evaluates the precision of explanations (Chen et al., 2024c; Limpijankit et al., 2025; Karvonen et al., 2026).
Reasoning faithfulness For the faithfulness of reasoning, methods improve the causal consistency of reasoning traces through interventions (Uikey, 2025), preference optimization over intervened reasoning chains Swaroop et al. (2025), multi-agent causal verification (Tang et al., 2025), and SCM guided reinforcement learning for faithful multi-hop reasoning (Bu et al., 2026).

4.3 Interpretability

Interpretability aims to explain the mechanisms behind LLMs behavior (Feldhus and Kopf, 2025; Zhang and Nanda, 2024). Causal approaches contribute to the field by identifying causally influential components in the model, distinguishing correlational associations from causal mechanisms (Meng et al., 2022).
Instance-Based Instance-based methods use counterfactual generation and augmentation to assess the prediction sensitivity and explain individual output (Han et al., 2020; Gat et al., 2024; Bhattacharjee et al., 2024; Nussbaum-Hoffer et al., 2026).
Model-Based Model-based methods use causal abstraction (Geiger et al., 2025; Sutter et al., 2025; Geiger et al., 2024; Wu et al., 2023), causal attribution (Syed et al., 2024). Alternatively, these methods focus on different levels such as feature-level (Ji et al., 2025; Bonagiri et al., 2026; Shah, 2026; Goyal et al., 2025; Wang et al., 2025c), component-level (Meng et al., 2022; Yan et al., 2026; Ortu et al., 2024), and system-level (Gupta et al., 2025a; Lyu et al., 2026; Triantafyllou et al., 2025; Hooda et al., 2024; Gupta et al., 2025b).

4.4 Robustness

Robustness is a crucial aspect for LLMs to maintain reliable performance despite external influences or distribution shifts (Wang et al., 2023; Zhu et al., 2024). In this section we discuss the robustness of responses and reasoning of LLMs.
Output robustness To enforce output robustness, training-based methods address distribution shifts and spurious correlations through SCM guided representation decomposition (Yu et al., 2026a) and causal identification of model components for targeted policy optimization (Zhou et al., 2025).
Reasoning robustness For robustness of reasoning, prompt-based methods leverage front-door-adjustment criteria (Zhang et al., 2025a; Wu et al., 2024b; Zhao et al., 2025; Li et al., 2026a), while training-based methods use counterfactual rewards (Wang et al., 2026a), counterfactual distillation of multi-view reasoning traces (Feng et al., 2024) and independent causal mechanisms that separate invariant and domain-specific representations of OOD generalization Gendron et al. (2024a).

4.5 Unlearning

Unlearning is the task of removing specific knowledge from an LLM, this is done by selectively removing the influence of data from the model while preserving its capabilities (Yao et al., 2024; Liu et al., 2024e). Causal approaches support unlearning through process modeling and model editing.
Modeling Modeling based methods rely on SCMs to mitigate confounding effects through backdoor adjustment (Liu et al., 2024d) or counterfactual reasoning traces with preference optimization to suppress targeted knowledge in large reasoning models (Li et al., 2026c).
Editing Model editing based approaches localize knowledge-bearing components through causal tracing and intervene on their activations for targeted removal (Fang et al., 2026). Another approach uses probabilities of sufficiency and necessity to guide attention-head interventions and fine-tuning to reduce LLM-toxicity (Wang et al., 2026d).

4.6 Security and Alignment

Security helps protect LLM or LLM-based systems from attacks, misuse, or other threats (Das et al., 2025), while, alignment ensures that the behavior presented by LLM is consistent with human intentions, preferences, and values (Shen et al., 2023). Causal approaches contribute to security by adversarial attacks, detection, and defense.

Attacks For attacks, front-door adjustment has been used to bypass LLM safety mechanisms through mediator-based jailbreak (Zhou et al., 2026).

Detection Detection methods use counterfactual probing to identify prompt injection vulnerabilities (Liu et al., 2025a), counterfactual information-flow analysis to trace unsafe agent propagations (Cai et al., 2026), interventional behavior mapping to detect LLM misbehavior (Zhang et al., 2025c), and causal influence diagrams to support proactive identification of harmful actions (Hahm et al., 2025).

Defense Defensive method use front-door adjustment with guard mediators to mitigate backdoor attack (Liu et al., 2024c), alongside counterfactual execution testing (He et al., 2026b), and temporal causal analysis (Zhang et al., 2026a) to prevent indirect prompt injection attacks.

Alignment For alignment, SCM based interventions enable counterfactual generation for pluralistic value alignment (Guo et al., 2025a), while causal reward modeling supports preference learning (Kobalczyk and Van Der Schaar, 2025).

5 Causality and LLM Applications

We discuss how the integration of causality and LLMs is applied to tasks in causal and data analysis, scientific discovery, and software engineering.

Traditional Causal Analysis focuses on causal reasoning using tabular or numerical data. LLMs support causal discovery by extracting variable metadata and domain knowledge (Liu et al., 2024a; Abdulaal et al., 2024; Li and Russo, 2026; Kim, 2026; Li et al., 2024a), generating causal priors (Lee et al., 2024; Kampani et al., 2024; Ban et al., 2025a,b; Baldo et al., 2026; Lee et al., 2025), or selecting between numerical methods and expert recommendations (Clivio et al., 2025). They also refine causal graphs using world knowledge (Khatibi et al., 2025; TAKAYAMA et al., 2025) or fine-tuning (Dong et al., 2025). Other studies integrate LLMs with traditional methods for iterative graph refinement (Du et al., 2025; Feng et al., 2025a), causal prior generation and refinement (Jiang et al., 2024; Ashwani et al., 2024; Dhawan et al., 2024; Sun et al., 2025b; Gendron et al., 2024b), and joint graph construction (Liu et al., 2026). For fairness-aware discovery, LLM-GUIDED CD (Zanna and Sano, 2025) combines LLM-based causal discovery (Jiralerspong et al., 2024) with active learning to identify fairness-relevant causal paths. Beyond causal discovery, LLMs serve as copilots for end-to-end causal reasoning (Berrevoets et al., 2025; Wang et al., 2025b; Verma et al., 2025) and construct executable SCMs for causal simulation (Astorga et al., 2026).

We cover the applications of data analysis, scientific discovery and software engineering in Appendix E

6 Evaluation and Benchmarking

This section presents an overview of the evaluation strategies for LLM-architecture, LLM-capability, LLM-trustworthiness, and applications. A comprehensive list of benchmarks can found in Table 1. Detailed discussion can be found in Appendix F.

The evaluation strategies mainly revolve around LLMs, in which they evaluate the causal effectiveness of post-training strategies (eg: CAUSALGYM (Arora et al., 2024)), evaluating reasoning (eg: RECITE (Saklad et al., 2026)) and generation (eg: CLOMO (Huang et al., 2024d)) aspects, trustworthiness of LLMs (eg: MIB (Mueller et al., 2025)). Whereas, the application benchmarks (eg: CAUSCIBENCH (Acharya et al., 2025)) evaluate the downstream capabilities of LLM.

7 Challenges & Future Works

Research at the intersection of causality and LLMs has advanced system design,model capabilities, and trustworthiness. Despite this progress, several challenges and open research directions persist.

Gap Between Causal Capabilities and Applications Although many LLM applications rely on causal inference and counterfactual reasoning, limited research directly addresses improving these capabilities. This gap calls for methods that strengthen causal reasoning and generalize across tasks and domains.

Causal Reliability of LLM Frameworks Many LLM-based frameworks, particularly in high-stakes applications, rely on generated explanations and reasoning whose faithfulness and causal validity are uncertain. Future research should develop causal evaluation methods to assess how these frameworks make decisions and whether their outputs are consistent with the underlying causal relationships.

Causal Validity and Generalization Existing evaluations often emphasize task accuracy on narrow or synthetic benchmarks, which may not adequately assess causal validity or generalization. Developing scalable evaluation methods across diverse tasks and real-world scenarios is an important research direction.

8 Conclusion

In this survey, we comprehensively explore the synergy of causality and LLMs. We present a detailed overview of the developing field and propose a taxonomy to highlight the various contributions through different perspectives. Although there are many developments in this field, we propose current challenges and future work to encourage the development of advanced LLMs and causality.

Limitations

This work aims to provide a comprehensive overview on the synergy of LLMs and causality. Our analysis is focused on the comprehensive breadth of LLM and causality encompassing both LLM of causality and causality for LLM perspectives hence, does not delve into the depths of internal mechanisms or implementation of causality or role of LLM in every work. Moreover, the field of causality and LLMs is a rapidly evolving research frontier where significant developments appear across NLP/ML venues and pre-prints. Given the pace and nature of this field our selection criteria prioritizes comprehensive coverage of key developments over exhaustive enumeration. For pre-prints, the authors of this paper manually reviewed them to assess their quality and relevance to the topic.

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A Causality Background

In this section, we give an overview of the background in causal inference. We introduce the key concepts and notations used throughout this survey. First, we define causal graphs and structural causal models, following the framework of Pearl (2009), then we move to define individual, conditional and average treatment effects using the potential outcomes framework of Rubin (1974). Finally, we define counterfactual inference, which concerns the question of what would have happened under a different intervention given what was actually observed.

Definition 1 (Causal Graph). A causal graph is a directed acyclic graph (DAG) $G = (V, E)$, where $V = \{V_1, \ldots, V_d\}$ is a set of random variables and $E \subseteq V \times V$ is a set of directed edges representing direct causal relationships. An edge $V_i \to V_j$ indicates that V_i is a direct cause of V_j relative to the variables represented in G. The parents of V_i, denoted by $\mathrm{Pa}_G(V_i)$, are the variables that directly influence V_i.

Definition 2 (Structural Causal Model). A structural causal model (SCM) is a tuple $\mathcal{M} = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P_{\mathbf{U}})$, where $\mathbf{U}$ denotes exogenous variables, $\mathbf{V} = \{V_1, \ldots, V_d\}$ denotes endogenous variables, $\mathcal{F} = \{f_1, \ldots, f_d\}$ is a collection of structural functions, and $P_{\mathbf{U}}$ is a joint probability distribution over the exogenous variables. Each endogenous variable V_i is determined by a structural equation

$$V_i = f_i\big(\mathrm{Pa}_G(V_i), U_i\big),$$

where U_i denotes the exogenous inputs to f_i. The structural equations specify the causal mechanisms generating the endogenous variables. An intervention $\mathrm{do}(T = t)$ replaces the structural equation for T by $T = t$, leaving the remaining structural equations unchanged.

Definition 3 (Individual Treatment Effect). Let $T \in \{0, 1\}$ denote a binary treatment and let $Y_i(t)$ denote the potential outcome of individual i under intervention $T = t$. The individual treatment effect (ITE) is defined as

$$\tau_i = Y_i(1) - Y_i(0).$$

The ITE quantifies the causal effect of changing the treatment from 0 to 1 for individual i. Since only one potential outcome is observed for each individual, the ITE is generally not directly identifiable from observed data.

Definition 4 (Conditional Average Treatment Effect). Let X denote a vector of pretreatment covariates. The conditional average treatment effect (CATE) at $X = x$ is defined as

$$\tau(x) = \mathbb{E}\big[Y(1) - Y(0) \mid X = x\big].$$

Equivalently,

$$\tau(x) = \mathbb{E}[Y(1) \mid X = x] - \mathbb{E}[Y(0) \mid X = x].$$

The CATE quantifies the average causal effect among individuals whose covariates satisfy $X = x$, allowing treatment effects to vary across covariate-defined subpopulations.

Definition 5 (Average Treatment Effect). The average treatment effect (ATE) is the expected difference between the potential outcomes under treatment and control:

$$\mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)].$$

Equivalently,

$$\mathrm{ATE} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}_X[\tau(X)].$$

The ATE quantifies the average causal effect of the treatment across the target population.

Definition 6 (Counterfactual Inference). Let $\mathcal{M}$ be a structural causal model, let $E = e$ denote observed evidence, and let Y_t denote the outcome that would result from the hypothetical intervention $\mathrm{do}(T = t)$. Counterfactual inference, in Pearl's structural causal framework, consists of evaluating

$$P_{\mathcal{M}}(Y_t = y \mid E = e).$$

This quantity represents the probability that the outcome would equal y under the hypothetical intervention $T = t$, conditional on the observed evidence $E = e$.

Counterfactual inference follows three steps:

1. **Abduction:** Update the distribution of exogenous variables using the evidence,

$$P_{\mathbf{U}}(\mathbf{u} \mid E = e).$$

2. **Action:** Construct the intervened model $\mathcal{M}_t$ by replacing the structural equation for T by $T = t$.

3. **Prediction:** Compute the distribution of Y in $\mathcal{M}_t$ under the updated distribution of the exogenous variables:

$$\begin{aligned} &P_{\mathcal{M}}(Y_t = y \mid E = e) \\ &= \int \mathbb{I}\{Y_{\mathcal{M}_t}(\mathbf{u}) = y\}\, dP_{\mathbf{U}}(\mathbf{u} \mid E = e). \end{aligned}$$

   assuming the structural equations admit a unique solution.

Unlike interventional inference, counterfactual inference conditions on factual evidence before evaluating a hypothetical intervention.

B Detailed Taxonomy

The main paper focuses on the individual topics of the relationship between causality and LLMs. In Figure 3 of this appendix shows the complete taxonomy. The taxonomy is organized into five main areas: causality and system design, causality and capability, causality and trustworthiness, applications, and evaluation. Each area is then divided into the topics covered in the main paper, and each topic is shown with the papers and sections where they are discussed.

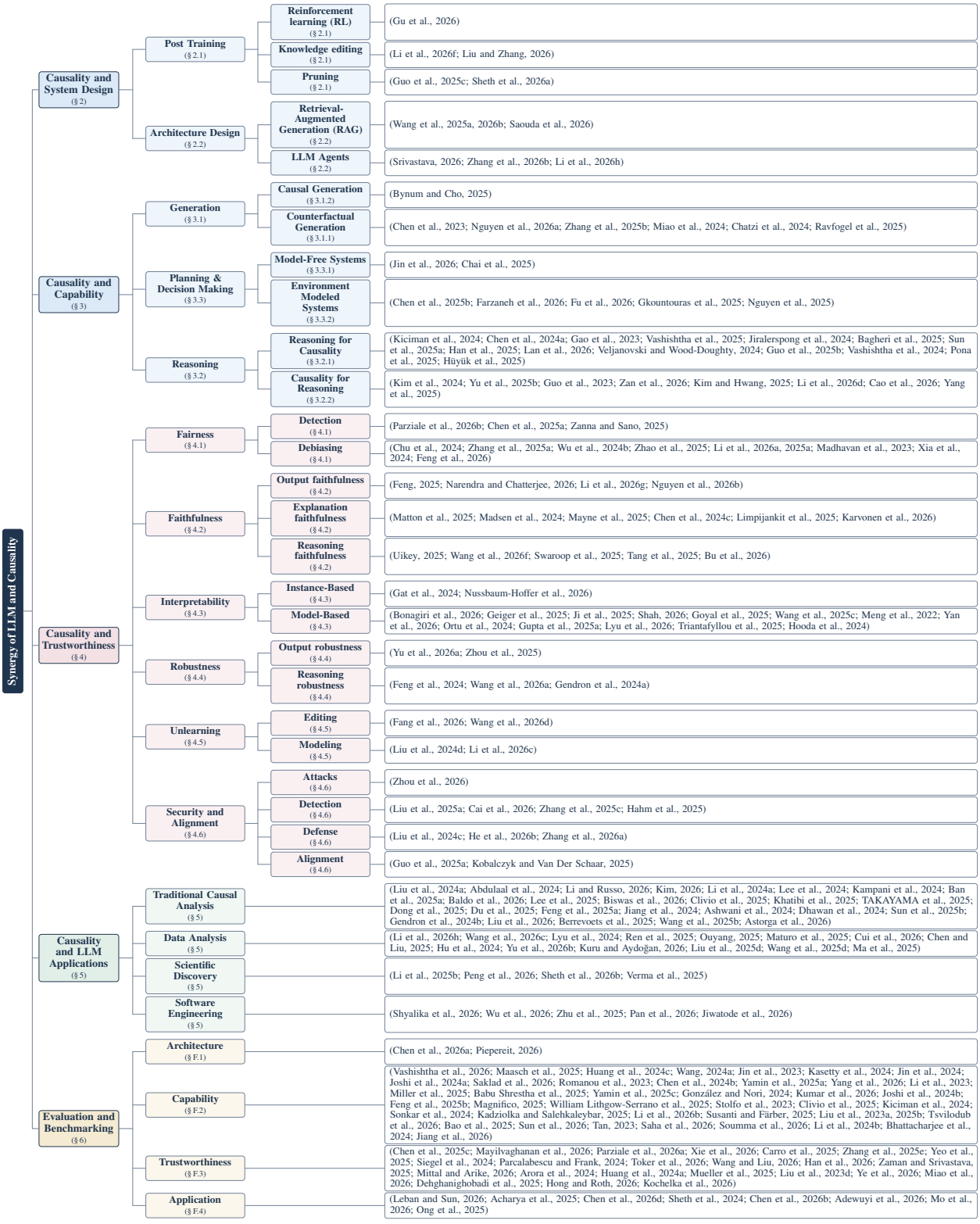


Figure 3: Taxonomy for synergy between LLM and causality.

C Detailed Comparison with Existing Surveys

Prior surveys on causality and LLMs typically address only one direction of the relationship: either how LLMs perform causal tasks, or how causal methods improve LLMs. Table 2 evaluates each survey against both directions and against the five areas of our taxonomy. We consider an area covered only when a survey discusses multiple works on that area in depth, rather than citing them in passing and under this criterion, no prior survey covers all five areas, which is the gap our survey addresses.

Study	Focus	LLM for Causality	Causality for LLMs	System Design	Capability	Trustworthiness	Apps.	Eval.
Zhang et al. (2023)	Causal feasibility and opportunities	⚠	✗	✗	✗	✗	✗	✗
Wu et al. (2024a)	Causality across the LLM lifecycle	⚠	✓	⚠	✓	✓	⚠	⚠
Cui et al. (2024)	Commonsense causality and reasoning	✗	✗	✗	✗	✗	✗	⚠
Ma (2025)	LLMs for causal inference	✓	✗	✗	⚠	✗	⚠	⚠
Liu et al. (2025c)	Causal–LLM collaboration, reasoning, trustworthiness, and multimodality	✓	✓	✗	✓	✓	⚠	⚠
Yu et al. (2025a)	Causal reasoning review and evaluation	✓	✗	✗	⚠	✗	⚠	⚠
Cui et al. (2025)	Causal uncertainty	⚠	✗	✗	⚠	✗	✗	✗
Bazgir et al. (2025)	Causal multi-agent LLM systems	✓	✗	✗	⚠	✗	⚠	⚠
Wan et al. (2025)	LLMs for causal discovery	✓	✗	✗	⚠	✗	⚠	✗
Frauen et al. (2026)	Causal LLM development and evaluation	✗	✓	⚠	✗	✗	✗	⚠
Li et al. (2025c)	Causal reasoning enhancement	✓	✗	✗	⚠	✗	⚠	⚠
Ours	**Causal LLM methods and evaluation**	✓	✓	✓	✓	✓	✓	✓

Table 2: Coverage of related surveys and studies, assessed against the taxonomy. In the five coverage areas, ✓: developed coverage of most branches of the area; ⚠: developed coverage limited to one or two branches; ✗: scattered citations or no developed coverage. **System Design**: causal methods for post-training and model design; **Capability**: generation, reasoning, and planning; **Trustworthiness**: fairness and bias, faithfulness, interpretability, robustness, unlearning, and security/alignment; **Apps.**: causal analysis, data analysis, scientific discovery, and software engineering; **Eval.**: Evaluation covers system design, capability, trustworthiness, and applications.

D Benchmarks, Datasets, and Metrics

Evaluation resources for causal reasoning in LLMs have been scattered across multiple independent works, making it hard to compare and evaluate them. Hence, we compile these resources into two tables, Table 3 and Table 1, to help readers compare them. The Table 3 lists benchmarks, datasets, and metrics by causal tasks. Readers can use this table to determine which resource is relevant to the causal task of interest, and what metrics are used to evaluate it. Table 1 regroups benchmarks and datasets by the LLM aspect they evaluate: system design, capability, trustworthiness, and applications. Each item in the tables cites the paper that introduced it.

E Application with More Details

In this section, we discuss in more detail the applications of LLM and causality. This is an expansion of our overview in Section 5. We cover the applications of data analysis, scientific discovery and software engineering.

Data Analysis Causal LLMs are used for various data analysis tasks. For anomaly detection they use the causal knowledge for tabular anomaly detection (Wang et al., 2026c). Causal LLMs are also used for analysis tasks involving sentiment analysis (Lyu et al., 2024; Ren et al., 2025; Ouyang, 2025), bankruptcy analysis (Maturo et al., 2025), time-series analysis (Cui et al., 2026; Biswas et al., 2026), sensory data analysis (Chen and Liu, 2025; Hu et al., 2024), and social emergence analysis (Yu et al., 2026b). Apart from analysis they are used in information retrieval (Jeong et al., 2026; Zhang and Oussalah, 2026; Naduvilakandy et al., 2025), recommendation systems (Kuru and Aydoğan, 2026; Liu et al., 2025d), question-answering (Wang et al., 2025d) and source fact-checking (Ma et al., 2025).

Scientific Discovery AI scientists in which AI systems, including LLMs, perform scientific discovery tasks have recently emerged. Recent studies have explored the causal capabilities of LLMs for experimental design (Li et al., 2025b), hypothesis generation (Peng et al., 2026), and instrumental variable discovery (Sheth et al., 2026b).

Software Engineering Causal LLMs are utilized to assist with software engineering tasks such as system analysis that uses causality identification methods to find key features in a programs (Shyalika et al., 2026; Wu et al., 2026), using causal reasoning for evaluation and optimization (Zhu et al., 2025; Pan et al., 2026), or using causal induction to reverse engineering video game description language (VGDL) rules from gameplay traces (Jiwatode et al., 2026).

F Evaluation with More Details

In this section, we discuss in more detail the evaluation of causal methods for LLMs. This is an expansion of our overview in Section 6. We cover the evaluation of system design, capability, trustworthiness, and applications as per our taxonomy. For each of them, we discuss the benchmarks and datasets that have been proposed or adapted, and how studies use interventions and counterfactuals to evaluate model behavior. The resources discussed in this section can be found in Table 3 and Table 1.

F.1 Architecture

To evaluate the impact of post-training methods on LLM's causal inference abilities CAUSALGYM (Chen et al., 2026a) evaluates the extent to which post-training methods enhance LLMs' capacity for causal inference. CausalGym comprises of seven core causal tasks for training and five diverse test sets. Using this dataset, they evaluate five post-training approaches: SFT, DPO, KTO, PPO, and GRPO.

For cross-model activation transfer, existing methods evaluate the representational similarity of the transferred activations and do not establish the causal influence (Piepereit, 2026). To bridge the gap, the authors intervene on the target model's hidden states injecting the source model's activations and a negative injection unrelated to the source model. They compare both the outputs to evaluate the causal effect of source model's activation transfer on the target model.

F.2 Capability

In this section, we highlight the different evaluation methods for evaluating the different capabilities of LLMs that are reasoning, generation, and planning.

To evaluate causal reasoning capabilities of LLMs some studies have relied on benchmarks and datasets that were originally designed for traditional causal tasks such as BNLearn (Scutari, 2010b) and Tübingen (Mooij et al., 2016). In some cases authors had to adapt these datasets to suit LLMs or their tasks (Kasetty et al., 2024; Clivio et al., 2025; Kiciman et al., 2024). However, various contributions have been made for datasets and benchmarks for causality in LLMs such as Cladder (Jin et al., 2023), Corr2Cause Jin et al. (2024),

COLD (Joshi et al., 2024a), CLEAR (Chen et al., 2024b), CRAB (Romanou et al., 2023), CausalBench (Wang, 2024b), RECITE (Saklad et al., 2026) that enable the evaluation of LLMs on different causal reasoning aspects such as formal, real-world, long-horizon, graph-based or certain novel tasks like MalAlgoQA (Sonkar et al., 2024) and CLOMO (Huang et al., 2024d). A comprehensive list of benchmarks can be found in Table 3.

A common focus while evaluating the LLMs for causal reasoning is to evaluate on the trustworthiness of their performed reasoning (Feng et al., 2025b; Kadziolka and Salehkaleybar, 2025; Kumar et al., 2026; William Lithgow-Serrano et al., 2025; Yang et al., 2026; Yamin et al., 2025c; Li et al., 2026b) and the extent of their reasoning capability (Vashishtha et al., 2026; Joshi et al., 2024b; Magnifico, 2025; Susanti and Färber, 2025; Miller et al., 2025; Li et al., 2023; Yamin et al., 2025b; Liu et al., 2023a; González and Nori, 2024; Liu et al., 2025b; Babu Shrestha et al., 2025).

Some evaluations for other reasoning capabilities have used causality as a medium for evaluation. For example, to evaluate the robustness of LLM in mathematical reasoning, a study develops a causal model of the the mathematical factors such as the surface form of the question, the operands, the math operators and the solution they study the behavior of LLMs to direct interventions in the input space (Stolfo et al., 2023). Similarly, to study how causal quantities compose across a graph or how the behavior changes when the composition or decomposition of causal measure change with graph complexity a study formalizes compositional causal reasoning CCR as the ability to infer the composition of local reasoning on global level and the decomposition of global level on local reasoning (Maasch et al., 2025). They define a framework which generates ground-truth for local components and its global composition and use that to evaluate the consistency of reasoning against ATE and PNS for composition and decomposition tasks. Alternatively, a study uses causal-mechanistic experiments to understand LLM's ToM abilities comprehensively (Tsvilodub et al., 2026).

Similarly, many studies have evaluated CoT reasoning with help of causality to understand its functioning and reliability (Bao et al., 2025; Sun et al., 2026; Tan, 2023; Saha et al., 2026).

Evaluations for generation capability of LLMs evaluate the extent of LLM's counterfactual generation on NLU tasks (Soumma et al., 2026; Li et al., 2024b), or data generation (Jiang et al., 2026).

A study investigates the effect of reasoning and semantic retrieval in CoT-reasoning (Kim et al., 2024). The authors compare the performance of LLM for reasoning questions based on RAG for retrieval-based setting, graph-inference for reasoning-based setting and proposed KG-based random-walk prompting framework for combination-setting. Their results reveal that incorporating causal relationships and structured reasoning can improve reasoning in LLMs.

F.3 Trustworthiness

A research highlights the lack of evaluating the spurious correlations in LLM semantics, existing studies detect them with co-occurrence statistics from training which are unavailable for LLMs. To assess the gap, LCD (Miao et al., 2026) identifies spurious correlations within the help of statistics encoded in the LLM parameters leveraging causal discovery. Based on these, they further introduce Spurious Correlation Evaluator (SCE) to assess the robustness using noisy data containing the identified spurious correlations.

It is important to evaluate the effectiveness of different methods to improve safety in LLMs for which researchers have evaluated different AI safety elements with a causal perspective.

Evaluation strategies in fairness focuses on counterfactual fairness (Parziale et al., 2026a; Mayilvaghanan et al., 2026), cognitive bias (Carro et al., 2025), social bias (Xie et al., 2026; Zhang et al., 2025e), and gender bias (Chen et al., 2025c). Faithfulness evaluations try to establish the trustworthiness of LLM reasoning based on concepts (Toker et al., 2026), SCE (Dehghanighobadi et al., 2025), CfSim (Hong and Roth, 2026), tabular reasoning (Wang and Liu, 2026), CoT reasoning (Parcalabescu and Frank, 2024) and large reasoning models (LRM) (Han et al., 2026). Similarly, interpretability evaluation focuses on understanding the effect of new methods causally on models architectures (Mueller et al., 2025; Arora et al., 2024; Huang et al., 2024a). Robustness and security rely on counterfactual generation or counterfactual analysis to establish the performance of LLMs in novel scenarios (Kochelka et al., 2026; Ye et al., 2026; Liu et al., 2023d).

F.4 Application

To understand the behavior of LLMs researchers have used causality as a medium for their evaluations. With the rising adoption of LLMs as AI scientists, researchers have proposed benchmarks to evaluate LLMs for their causal capabilities as Causal AI Scientists. Some studies focus on the causal reasoning capability of AI Scientists (Leban and Sun, 2026; Acharya et al., 2025; Chen et al., 2026d), while others propose novel tasks (Sheth et al., 2024). To ensure safe LLM reasoning for medical recommendations, studies have leveraged counterfactuals to generate benchmarks and datasets focused on robustness (Chen et al., 2026b), fairness (Adewuyi et al., 2026), and faithfulness (Mo et al., 2026).

Alternatively, a study investigated the use of forward counterfactual generation as a means to benchmark LLMs to anticipate future market developments (Ong et al., 2025).

Task	Benchmarks	Datasets	Metrics
Causal Reasoning	CLadder (Jin et al., 2023); CLEAR (Chen et al., 2024b); CausalGraph2LLM (Sheth et al., 2025); CausalEval (Yu et al., 2025a); METER (Li et al., 2026e); CausalPitfalls (Du et al., 2026); Causal Chains, Natural Word Chain, and Intuitive Physics tasks (Zečević et al., 2023); CausalDS (Leban and Sun, 2026); CauSciBench (Acharya et al., 2025); CauGym (Chen et al., 2026a); CaLM (Chen et al., 2024a); NoisyCausal (Xu and Fu, 2026); CausalProbe-2024 (Chi et al., 2024); AC-Bench (Zhang et al., 2025d); CRAB (Romanou et al., 2023); CausalBench (Wang, 2024a); CausalFlip (Wang et al., 2026e); QRData (Liu et al., 2024b)	ExpliCa (Miliani et al., 2025); Corr2Cause (Jin et al., 2024); CauSim (Astorga et al., 2026); QRText (Liu et al., 2024b); CausalityCheck-CP (Li et al., 2026b); COPA (Roemmele et al., 2011); e-CARE (Du et al., 2022); WIQA (Tandon et al., 2019); CommonWhy (Toroghi et al., 2026)	DoVerifier (He et al., 2026a)
Causal Inference	CausalReasoning Benchmark (Sawarni et al., 2026); Missing-causal-variable benchmark (Sheth et al., 2024)	CandyParty (González and Nori, 2024); Null-contingency medical scenarios (Carro et al., 2025)	ATE error (Rubin, 1974); CATE / ITE error (Shalit et al., 2017); PEHE (Hill, 2011) / standardized PEHE (Bynum and Cho, 2025); Probability of Necessity and Sufficiency (Tian and Pearl, 2000); Probability of Necessity (Tian and Pearl, 2000); Probability of Sufficiency (Tian and Pearl, 2000); Compositional ATE / PNS error (Maasch et al., 2025); Causal identification accuracy (Sawarni et al., 2026); Causal estimation error (Sawarni et al., 2026); Mediation Influence Score (Sheth et al., 2024)
Causal Discovery	CausalGraphBench (Babakov et al., 2025); Latent-Confounder benchmark (Ding et al., 2026)	bnlearn Bayesian-network repository (Scutari, 2010a); Sachs (Sachs et al., 2005); Tübingen Cause–Effect Pairs (Mooij et al., 2016); ER and scale-free synthetic DAGs (Erdős and Rényi, 1959; Barabási and Albert, 1999)	Structural Hamming Distance (Tsamardinos et al., 2006); Normalized Hamming Distance (Kumar et al., 2026); Structural Intervention Distance (Peters and Bühlmann, 2015); Adjacency precision / recall / F1; Edge FPR / FDR / TPR; Edge AUROC / AUPRC; Causal-edge calibration (Kumar et al., 2026); Causal DAG construction score (Li et al., 2026g)

Continued on the next page

Table 3 continued from the previous page

Task	Benchmarks	Datasets	Metrics
Counterfactual Reasoning	CounterBench (Chen et al., 2026c); CofCA (Wu et al., 2025); CounterLogic (Joishy et al., 2026); MalAlgoQA (Sonkar et al., 2024); RECALL (Liu et al., 2023c); ContraTable (Wang and Liu, 2026); RFEval (Han et al., 2026); MamaBench (Adewuyi et al., 2026); MedEinst (Chen et al., 2026b); MEDCOUNTERFACT (Mo et al., 2026)	CRASS (Frohberg and Binder, 2022); Executable-Code Counterfactuals (Vashishtha et al., 2026); Counterfactual-simulatability sets (Hong and Roth, 2026); DailyDilemmas (Chiu et al., 2025)	Counterfactual-query accuracy; Factual / counterfactual inconsistency (González and Nori, 2024); Algorithm / malgorithm identification (Sonkar et al., 2024); Counterfactual performance gap; Bias Trap Rate (Chen et al., 2026b); Counterfactual evidence adherence (Mo et al., 2026)
Counterfactual Generation	CEval (Nguyen et al., 2024); FIN-FORCE (Ong et al., 2025); CHIVE (Karvonen et al., 2026)	CLOMO (Huang et al., 2024c)	Counterfactual validity / target success (Wachter et al., 2017); Proximity / minimality (Wachter et al., 2017); Plausibility / naturalness; Diversity (Mothilal et al., 2020); Self-Evaluation Score (Huang et al., 2024c); Excess Gower distance (Mayne et al., 2025)
Causal Generation	SD-SCM causal-effect benchmark (Bynum and Cho, 2025)		
Planning & Decision Making	CausalGame (Chen et al., 2026d)	SymFail (Luan et al., 2026); CAR planted-SCM cases (Shah, 2026)	Failure reproduction rate (Luan et al., 2026); Conditional causal-repair rate (Luan et al., 2026); Single-step contrastive causal effect (Shah, 2026); Shapley causal-attribution recovery (Shah, 2026)

Table 3: Benchmarks, datasets, and metrics for evaluating the causal abilities of large language models, grouped by causal task. A benchmark is a task suite with a fixed protocol, a dataset is a collection of examples for training or testing, and a metric is a score for model outputs. Each entry cites the paper that introduced it.